

1. A 50 kV direct-current power line consists of two wire conductors 2 m apart. When this line is transmitting 10 MW of power, how strong is the magnetic field midway between the conductors?
2. Consider two circular rings of copper wire. One ring is a scaled-up version of the other, twice as large in all regards (radius, crosssectional radius). If currents around the rings are driven by equal voltage sources, how do the magnetic fields at the centers compare?
3. (a) Find $\mathbf{H}$ in rectangular components at $P(2, 3, 4)$ if there is a current filament on the z axis carrying 8 mA in the $\hat{a}_z$ direction. (b) Repeat if the filament is located at $x = 1, y = 2$. (c) Find $\mathbf{H}$ if both filaments are present. (Hint: $\vec{B} = \mu_0 \vec{H}$)
4. The parallel filamentary conductors shown in Figure 1 lie in free space. Plot $|\mathbf{H}|$ versus y , $4 < y < 4$, along the line $x = 0, z = 2$.

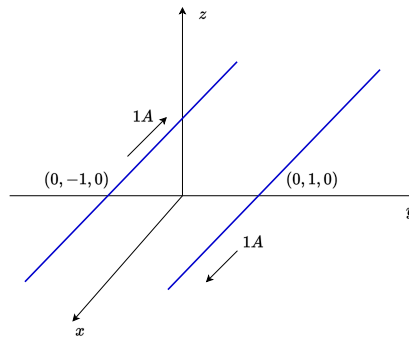


Figure 1: Q4

5. An infinite filament on the z axis carries 20π mA in the $\hat{a}_z$ direction. Three $\hat{a}_z$ -directed uniform cylindrical current sheets are also present: 400 mA/m at $\rho = 1$ cm, 250 mA/m at $\rho = 2$ cm, and 300 mA/m at $\rho = 3$ cm. Calculate H_ϕ at $\rho = 0.5, 1.5, 2.5$, and 3.5 cm.
6. A square wire frame with side length ℓ has total resistance R . It is being pulled with speed v out of a region where there is a uniform magnetic field $\mathbf{B}$ pointing out of the page (the shaded area in Fig. 2). Consider the moment when the left corner of the square is a distance x inside the shaded area.
 - (a) What force do you need to apply to the square so that it moves with constant speed v ?
 - (b) Verify that the work you do from $x = x_0$ (which you can assume is less than $\ell/\sqrt{2}$) down to $x = 0$ equals the energy dissipated in the resistor.

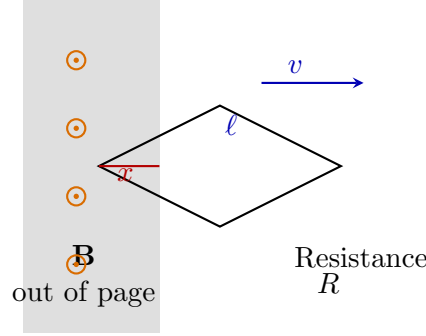


Figure 2: A square conducting frame being pulled out of a region of uniform magnetic field.

7. A metal crossbar of mass m slides without friction on two long parallel conducting rails, separated by a distance b , as shown in Fig. 3. A resistor R is connected across the rails at one end; compared with R , the resistance of the bar and rails is negligible. There is a uniform magnetic field $\mathbf{B}$ perpendicular to the plane of the figure. At time $t = 0$ the crossbar is given a velocity v_0 toward the right.

What happens afterward?

- Does the rod ever stop moving? If so, when?
- How far does it go?
- How about conservation of energy?

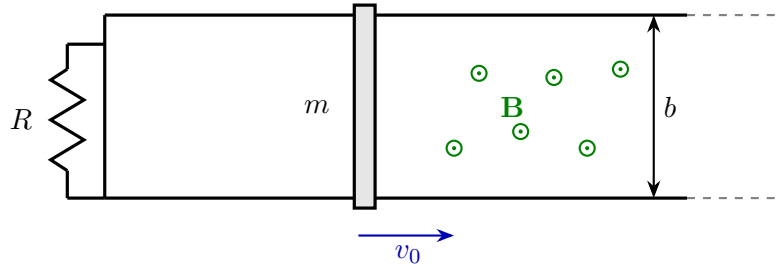


Figure 3: Sliding conducting bar on parallel rails in a uniform magnetic field.

8. A current I flows along a wire toward a point charge, causing the charge to increase with time. Consider a spherical surface S centered at the charge, with a tiny hole where the wire enters, as shown in Fig. 4. The circumference C of this hole is the boundary of the surface S .

Verify that the integral form of Maxwell's equation,

$$\oint_C \mathbf{B} \cdot d\mathbf{s} = \iint_S \left(\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \mu_0 \mathbf{J} \right) \cdot d\mathbf{a}, \quad (1)$$

is satisfied.

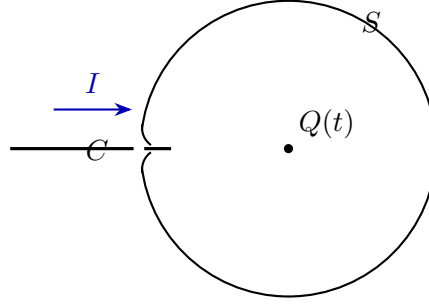


Figure 4: A spherical surface with a small hole through which the current-carrying wire enters.

9. Consider the magnetic flux density $\vec{B} = \frac{\mu_0 I}{2\pi r} \hat{\phi}$ produced by the current I flowing along the x -axis.

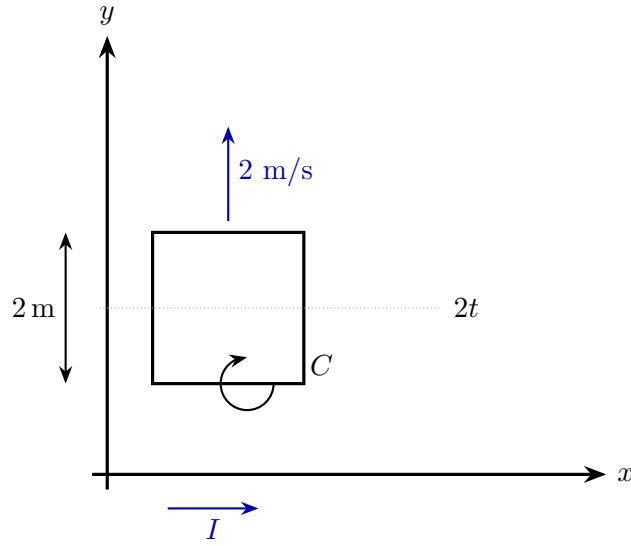


Figure 5: Square loop of area 4 m^2 moving upward in the magnetic field produced by a current-carrying wire along the x -axis.

What is the emf $\mathcal{E}$ of a square loop C of area 4 m^2 moving on the xy -plane with edges parallel to the x - and y -axes, if its center is located at $y = 2t$ m as a function of time? Compute the emf $\mathcal{E}$ first as

$$-\frac{d\Phi}{dt}$$

and then as

$$\oint (\vec{v} \times \vec{B}) \cdot d\vec{l}$$

to verify that the same values are obtained.